WORKED SOLUTION

# STEP I 2005, Question 5

Pure • Integrating Powers

**STEP I 2005, Question 5**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q5

## The key idea

Powers of $x+1$ integrate by the reverse chain rule, $\int (x+1)^n \,\mathrm{d}x = \frac{(x+1)^{n+1}}{n+1}$, with one exception: when $n=-1$ the answer is a logarithm. So every case split here comes from asking "when does the power become $-1$?". In (ii) the extra factor $x$ is removed by writing $x=(x+1)-1$, which leaves two plain powers and so *two* exceptional values of $m$.

## Solution

### Part (i)

If $k \neq 0$, the power $k-1$ is not $-1$, so

$$\int_0^1 (x+1)^{k-1}\,\mathrm{d}x = \left[\frac{(x+1)^k}{k}\right]_0^1 = \frac{2^k-1}{k}.$$

If $k=0$, the integrand is $\frac{1}{x+1}$, and since $x+1>0$ on $[0,1]$,

$$\int_0^1 \frac{1}{x+1}\,\mathrm{d}x = \Big[\ln(x+1)\Big]_0^1 = \ln 2.$$

**Ask yourself** Why should the answer for $k \neq 0$ have anything to do with the answer for $k=0$?

Because the *integrand* barely changes when $k$ moves a little away from 0. For $0 \le x \le 1$,

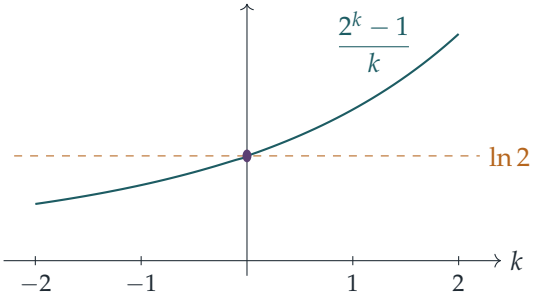
$$\left|(x+1)^{k-1} - (x+1)^{-1}\right| = \frac{\left|(x+1)^k - 1\right|}{x+1} \le \left|2^k - 1\right|,$$

since $1 \le x+1 \le 2$ means $(x+1)^k$ lies between 1 and $2^k$. Integrating over an interval of length 1, the two integrals differ by at most $|2^k-1|$, which tends to 0 as $k \to 0$. Hence,

for $k \approx 0$ (and $k \neq 0$),

$$\frac{2^k - 1}{k} = \int_0^1 (x+1)^{k-1}\,\mathrm{d}x \approx \int_0^1 (x+1)^{-1}\,\mathrm{d}x = \ln 2,$$

as required. In other words, the formula for $k \neq 0$ has a "hole" at $k = 0$, and the $k = 0$ integral fills it in exactly.




**Part (ii)**

Write $x = (x+1) - 1$, so that

$$x(x+1)^m = (x+1)^{m+1} - (x+1)^m.$$

Each term is a power of $x+1$, and integrating it produces a logarithm exactly when its power is $-1$. That happens for $m+1 = -1$ or $m = -1$, so the cases are $m = -1$, $m = -2$, and everything else.

**General case, $m \neq -1, -2$.**

$$\int_0^1 x(x+1)^m\,\mathrm{d}x = \left[\frac{(x+1)^{m+2}}{m+2} - \frac{(x+1)^{m+1}}{m+1}\right]_0^1 = \frac{2^{m+2}-1}{m+2} - \frac{2^{m+1}-1}{m+1}.$$

Over a common denominator the numerator is

$$(m+1)(2^{m+2}-1) - (m+2)(2^{m+1}-1) = 2^{m+1}\big(2(m+1) - (m+2)\big) + 1 = m\,2^{m+1} + 1,$$

so

$$\int_0^1 x(x+1)^m\,\mathrm{d}x = \frac{m\,2^{m+1}+1}{(m+1)(m+2)}, \qquad m \neq -1, -2.$$

**Case $m = -1$.**

$$\int_0^1 \left(1 - \frac{1}{x+1}\right)\mathrm{d}x = \Big[x - \ln(x+1)\Big]_0^1 = 1 - \ln 2.$$

**Case $m = -2$.**

$$\int_0^1 \left(\frac{1}{x+1} - \frac{1}{(x+1)^2}\right)\mathrm{d}x = \left[\ln(x+1) + \frac{1}{x+1}\right]_0^1 = \ln 2 + \tfrac{1}{2} - 1 = \ln 2 - \tfrac{1}{2}.$$

**Ask yourself** The general formula has $0/0$ at $m = -1$ and $m = -2$. Is that a contradiction?

No: it is part (i) again. At $m = -1$ the numerator is $-1 \cdot 2^0 + 1 = 0$, and at $m = -2$ it is $-2 \cdot 2^{-1} + 1 = 0$, so the formula is undefined there rather than wrong, and as in part (i) its limiting values are the special-case answers ($1 - \ln 2 \approx 0.307$ and $\ln 2 - \frac{1}{2} \approx 0.193$).

## Checks

- $m = 0$: $\int_0^1 x \, dx = \frac{1}{2}$, and the formula gives $\frac{0+1}{1 \cdot 2} = \frac{1}{2}$. For $m = 1$: $\int_0^1 (x^2 + x) \, dx = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$, and the formula gives $\frac{4+1}{6}$.
- Numerically, $k = 10^{-4}$ gives $\frac{2^k - 1}{k} \approx 0.693171$, against $\ln 2 \approx 0.693147$.
- Simpson's rule on all the integrals, for many values of $k$ and $m$ (including $m = -1, -2$), agrees with the formulas to $10^{-9}$. Putting $m = -1 \pm 10^{-5}$ into the general formula gives $0.30685\ldots$, matching $1 - \ln 2$.

## Common slips

- Writing $\frac{(x+1)^0}{0}$ for $k = 0$, or missing that (ii) has *two* exceptional values of $m$, not one.
- Integrating by parts in (ii) and still dividing by $m + 1$ without noticing that this fails when $m = -1$.
- In the $m = -2$ case, getting the sign of $\int (x+1)^{-2} \, dx = -(x+1)^{-1}$ wrong.

## Beyond STEP

The deduction in (i) says $\lim_{k \to 0} \frac{2^k - 1}{k} = \ln 2$, which is the derivative of $2^k$ at $k = 0$. The general fact used is that if integrands converge *uniformly*, their integrals converge too. That is a central theorem of first-year analysis, and it fails without uniformity. The same idea explains why $\int x^n \, dx$ "becomes" $\ln x$ as $n \to -1$. Abbott's *Understanding Analysis* treats uniform convergence and integration carefully.

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